Walkthrough: Build & Deploy a REST API

We will create a **`students-api`** Cloud Function with two endpoints: - `GET /students` - `POST /students`

Step 1 — Create a project directory

```
mkdir students-api
```

Now open the directory in the build-in code editor.

File -> Open Folder -> Type the directory path

Step 2 — Connect Editor Session to Google Cloud Project

Set the project in the terminal by running the following command:

```
gcloud config set project <your-project-id>
```

Step 3 — Create the Code File

NodeJS

1. Initialize the project

   ```
   npm init -y
   ```

2. Install the project dependencies

   ```
   npm install express
   ```

3. Create an `index.js` file

index.js

```
// index.js
const express = require("express");
const app = express();

app.get("/students", (req, res) => {
    res.json({ message: "Hello from Students API!" });
});

exports.studentsApi = app;
```

Python

main.py

```
def campus_api(request):
    if request.method == 'GET':
        return "Hello from the Students API!", 200

    if request.method == 'POST':
        data = request.get_json(silent=True) or {}
        return {"received": data}, 200

    return ("Method Not Allowed", 405)
```

Step 4 — Start the REST API locally

NodeJS

1. Install testing package

```
npm install --save-dev @google-cloud/functions-framework
```

2. Modify `scripts` property in `package.json`

package.json

```
"scripts": {
    "test": "...",
    "start": "npx functions-framework --target=studentsApi
}
```

3. Start the local server

```
npm run start
```

Python

1. Install testing package

```
pip3 install functions-framework
```

2. Start the local server

```
functions-framework --target=students_api
```

Step 5 — Test the API

1. Download and Install Postman
2. Open Postman
3. Click the "New" button at the top left
4. Select HTTP request
5. In your shell editor, click "Web Preview" in the top right and select "Preview on port 8080"
6. Copy the URL in the address bar
7. Paste it in the URL bar of Postman
8. Click the "Send" button

Step 6 — Deploy to Google Cloud

1. In the terminal, run the following command

```
gcloud functions deploy students-api \
    --gen2 \
    --region=us-central1 \
    --runtime=nodejs22 \
    --source=. \
    --entry-point=studentsApi \
```

```
    --trigger-http \
    --allow-unauthenticated
```

2. Run this command in the terminal to get the function URI

```
 gcloud functions describe students-api --region=us-central1 --
format='value(serviceConfig.uri)'
```

3. Navigate to that URL and add "/students" to view the API response.